

SOC Analyst Career Pathway

Security Operations Centre (SOC) Analysts are the team within an organization that protects it against cyber threats. SOC analysts are categorized into different levels, called tiers, to make handling security problems easier and faster. They Tier 1, Tier 2, Tier 3 and they show increasing levels of skill and responsibility in cybersecurity tasks. Each tier has its own role, from watching for alerts to analysing serious threats and handling complex incidents. Understanding these roles is important for organizations that are building or improving their SOC.

This document explains the three main tiers of SOC Analysts, their missions, duties, tools, and challenges.

Tier 1: Monitoring and Triage

A Security Operations Centre (SOC) Analyst Tier 1 is the first line of defence in a company's cybersecurity team. Their main role is to watch for possible security threats, check if they are real, and pass them to higher tiers if needed.

Main Duties

- **Monitor alerts:** Watch the security tools and systems for anything suspicious.
- **Triage:** Decide if an alert is false or real.
- **Investigate:** Look at logs and simple data to learn more about the alert.
- **Escalation of threats:** Pass serious threats to Tier 2 or Tier 3 analysts.
- **Documentation:** Write down what happened and what actions were taken.
- **Following playbooks:** Use step-by-step guides for common issues like phishing emails or brute force attacks.

Example tasks

- Run KQL queries on failed logons, summarize results by account or IP, and link related alerts.
- Use Microsoft Defender for Endpoint (MDE) to isolate a host after checking policy.
- Run enrichment playbooks (OTX, VirusTotal) and add findings to the case.

Skills Needed

- Basic knowledge of computer networks.
- Understanding of common attacks such as malware, phishing, brute force, etc.
- Knowing how to use SIEM tools such as Microsoft Sentinel.
- Ability to read and understand system logs from Windows and Linux.
- Good communication skills to explain findings clearly.
- Patience and attention to detail.

Challenges

- Many alerts are false positives, which can be boring or stressful.
- A high number of alerts every day.
- Work can include night or weekend shifts because SOC's run 24/7.
- New threats and tools appear often, so learning never stops.

Therefore, Tier 1 SOC Analysts are essential in cybersecurity. They make sure that alerts are validated, documented, and escalated when needed. The work can be repetitive, but it builds the foundation for higher-level roles in investigation, response, and leadership.

Tier 2: Investigation and Response

A Tier 2 SOC Analyst is the next level after Tier 1. While Tier 1 focuses on watching and triaging alerts, Tier 2 goes deeper. Their main job is to investigate suspicious activity, respond to real incidents, and reduce the chance of the same problem happening again.

Main Duties

- **Investigating alerts:** They look deeper into suspicious activity found by Tier 1.
- **Responding to incidents:** Take steps to contain and fix problems, for example, disabling accounts and isolating affected machines.
- **Writing reports:** Document findings, actions, and lessons learned.
- **Adjust Security rules:** Adjust SIEM rules to cut down false positives and improve accuracy.
- **Guide Tier 1 analysts:** Check the quality of their triage and help them improve.
- Daily work includes running detailed queries, reviewing logs across systems, performing endpoint forensics, and handling evidence.

Example Tasks

- Write KQL queries that join data from SecurityEvent, AzureActivity, and SignInLogs.
- Disable an affected/compromised user account, revoke sessions, and reset the password.
- Come up with new analytics rules to reduce the number of false positives.
- Collect and review evidence from an incident to build a timeline of what happened.

Skills Needed

- Strong knowledge of networks, operating systems, and security concepts.
- Ability to perform forensic checks on endpoints.
- Comfortable writing and editing SIEM queries, e.g, KQL.
- Incident response skills such as containment, elimination, recovery.
- Good reporting and communication skills to share results with managers or other teams.
- Ability to mentor and support Tier 1 analysts.

Challenges

- Most Investigations can be complex and require patience.
- Handling high-priority incidents can be stressful.
- Analysts must balance between having fast response and handling evidence carefully.
- Keeping up with new attack techniques and updating detections regularly.

Therefore, Tier 2 SOC Analysts play a critical role by digging deeper into alerts and taking action against real threats. They also make systems stronger by creating better rules and helping Tier 1 analysts. This role is challenging but rewarding, as it builds the skills needed to become a senior analyst or SOC leader.

Tier 3: Hunting & Detection Engineering

A Security Operations Centre (SOC) Analyst Tier 3 is the senior analyst in the SOC team. While Tier 1 watches alerts and Tier 2 investigates incidents, Tier 3 goes further by hunting for hidden threats, building strong detection rules, and having advanced incident response. They are often seen as experts who guide the whole SOC.

Main Duties

- **Threat hunting:** Search for unusual or suspicious behaviour in systems, even without an alert.
- **Detection engineering:** Create and test new detection rules for the SIEM.
- **Advanced incident response:** Handle serious or complex attacks, such as ransomware or insider threats.
- **Testing and validation:** Run purple-team or red-team style tests to make sure detections work.
- **Mentorship:** Support Tier 1 and Tier 2 analysts by reviewing their work and teaching them advanced techniques.
- Daily work includes writing advanced queries, compare and match up data across different sources, and tune detections.

Example Tasks

- Hunt for credential dumping activity (stealing of credentials to perform malicious activities) and turn the findings into a detection rule.
- Deploy updated configurations to improve visibility and reduce false positives.
- Run a Red Team test to simulate an attack and check if detections trigger
- Lead a post-incident review and propose new use cases to prevent similar attacks.

Skills Needed

- Deep knowledge of cybersecurity, including malware, techniques, and advanced attacks.
- Strong query and scripting skills for building detections e.g., KQL, PowerShell, Python.
- Ability to perform advanced forensics and data correlation across many systems.
- Experience with frameworks like MITRE ATT&CK to map and track threats.
- Leadership and communication skills to guide the SOC team and report to management.

Challenges

- Hunting can produce lots of data, and finding real threats can be difficult.
- Keeping up with new attack techniques requires continuous learning.
- Pressure is high during major incidents, as Tier 3 analysts often lead the response.
- Balancing detection improvements with minimizing false positives.

Tier 3 SOC Analysts are seniors, defenders who focus on finding hidden threats and building strong defences. They lead during major incidents, improve detection systems, and train the rest of the team. This role is challenging but is the gateway to SOC leadership and specialized expert positions.

Conclusion

The SOC Analyst career pathway builds step by step:

- **Tier 1:** learns to monitor and triage.
- **Tier 2:** investigates and responds to threats.
- **Tier 3:** hunts for hidden threats and builds stronger defences.

Each tier is important for keeping an organization secure. Together, they form a strong team that protects against cyberattacks and improves security over time.